Trumen Guard

A collision warning that runs on any phone, before the crash

Mode: Partner product
Owned by Toyota Royal Motors and BYD Royal Motors after-sales; changes the vehicle handover

Try it on your phone now

trumen-com.github.io/road-vision

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Most cars on our roads would not warn before a crash

2,188
killed on Uzbekistan's roads in 2025

46%
of crashes hit a pedestrian

≈ $2 bn
yearly cost, about 3% of GDP

−27%
rear-end crashes with a warning

- Most of the 5.5 million registered vehicles have no forward collision warning: it comes only with new, higher trims, and aftermarket units cost hundreds of dollars installed.
- Almost every driver already carries a camera and a processor that can do it: their phone.
- Crashes into pedestrians and cyclists are more than half of all crashes here. A warning must cover people, not only cars.

Sources: Gazeta.uz (Jan 2026; Nov 2025), Asian Transport Observatory road safety profile 2025, IIHS (forward collision warning alone, rear-end crashes).

Royal Motors after-sales owns it; the handover changes

Driver

Mounts the phone, opens one web page. Hears a warning about the car ahead or a person stepping out. Gets a trip report. Pays nothing for the basic warning.

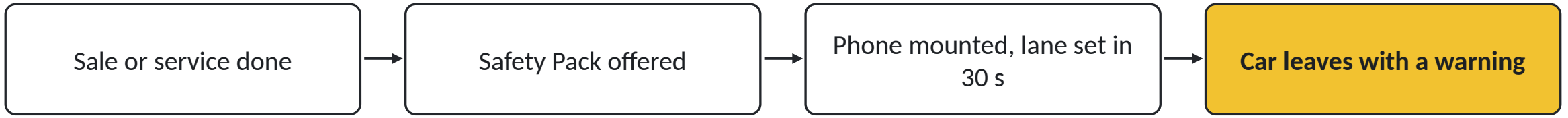
Royal Motors after-sales

Owns and sells the Safety Pack (mount, set-up, a year of trip reports) at handover and at service visits. Keeps the customer connected between visits.

Fleet customers

Taxi and corporate fleets buying Toyota and BYD in bulk pay per vehicle per month for warnings and near misses per driver, for coaching.

The process it changes: vehicle handover



Live demo

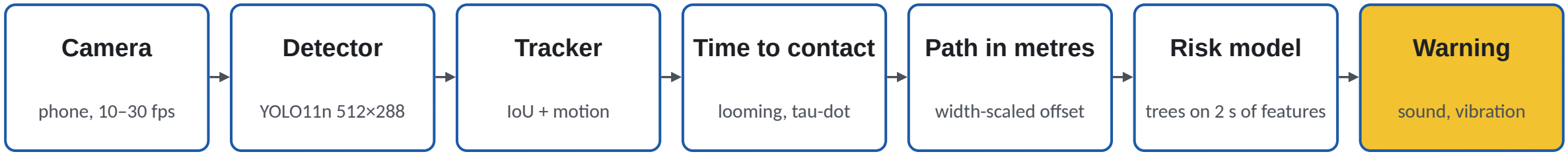
- A real dashcam crash clip: the band turns yellow, then red, with a beep, before impact
- Your video: open any dashcam file; it is analysed in real time in the browser
- Your camera: point a phone at the road or a street scene
- The trip report: every warning with what, where and time to contact; nothing leaves the phone



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Backup: a recorded run of the demo is in the repository (demo/backup.mp4).

Physics of looming plus a model learned on real crashes



All inside the phone browser (onnxruntime-web). No video leaves the phone.

Time to contact from one camera

An approaching car grows: d(ln h)/dt = 1/TTC. No distance or speed needed. Height, not width: a turning car gets wider, not taller.

Is it in my path, in metres

Offset ÷ the object's own width does not change with distance, so a parked car stays beside us and a car in our lane stays at zero.

Braking in control or a crash course

Tau-dot (Lee, 1976): when time to contact stops shrinking, the driver is handling it. Turning is removed using the whole scene's sideways motion.

Ours: tracker, looming fit, metric path test, tau-dot, features, risk model, alert logic, evaluation harness, app. Not ours: YOLO11n weights (AGPL-3.0), onnxruntime (MIT).

Tested on 667 held-out real dashcam clips

0.69
mAP, Nexar protocol

21%
of crash clips warned before the event

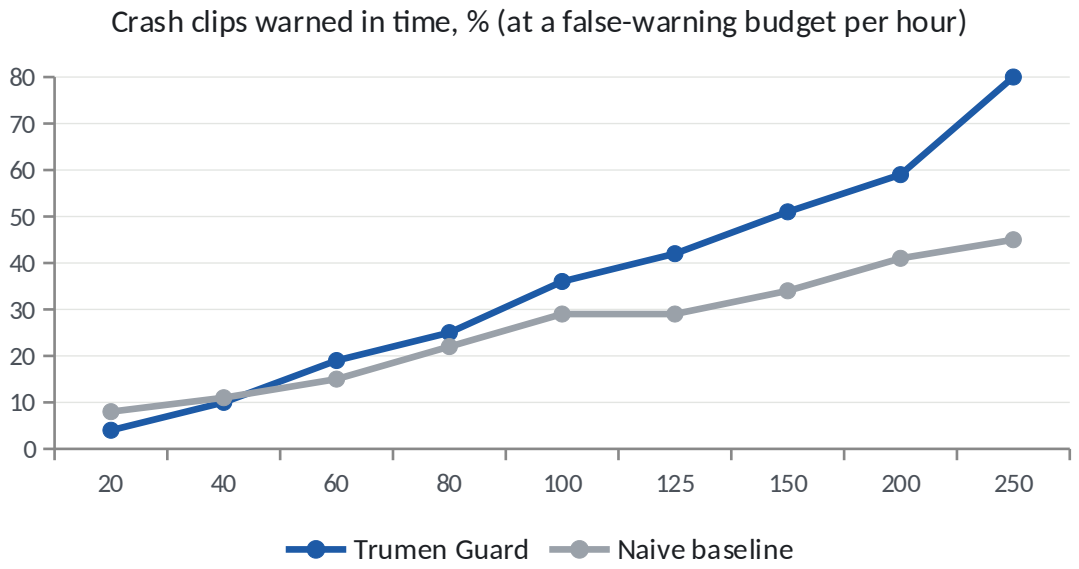
63
false warnings per hour of normal driving

0.72
ROC AUC

Version	mAP	AUC
Naive: biggest object in front	0.70	0.74
Tracking + time to contact + path, hand-set	0.65	0.67
Trumen Guard (learned trees on 2 s of features)	**0.69**	**0.72**
Tried, not shipped: + visual model (lowered mAP)	0.66	–

Ranking quality ties the naive baseline (mAP 0.69 vs 0.70). As a live warning, at the same false-warning rate, ours warns before more crashes: 19% vs 15% at 60 per hour, 51% vs 34% at 150.

The threshold was fixed on the training split before testing. mAP 95% interval 0.64–0.76; Kaggle winners reached about 0.9 with server video networks.



Night, rain, speed and cost, and where it fails

Condition	Clips	mAP	95% interval
Daylight	611	0.69	0.63–0.75
Twilight or dark	56	0.76	0.61–0.91
Clear or cloudy	635	0.71	0.66–0.77
Rain	32	0.58	0.44–0.82
Urban	315	0.71	0.64–0.79
Highway	183	0.78	0.70–0.87
Suburban, rural, other	169	0.61	0.51–0.75

Speed and cost

- Inference on the phone: WebGPU, or multi-threaded WebAssembly
- Detector 46 ms per frame on 2 CPU threads; the live frame rate shows on screen
- No server per car; static hosting under 1 cent per car per month
- Local cars and plates: classes, not brands; plates are never read

Known failure cases

- Crashes from the side or behind: one forward camera cannot see them
- Unlit pedestrians at night: the detector misses them
- Rain is our weakest group (32 clips, mAP 0.58): wipers and glare break the boxes
- Dense stop-and-go traffic: short time to contact is normal there; silenced below 8 km/h by GPS

Why another team cannot rebuild it in a weekend

Measured pipeline

Every rule and threshold was tuned against hundreds of real crash clips; the harness runs the exact browser code headless over the dataset.

Physics, not only a model

Looming, tau-dot and the metric path test explain every warning: which object, where, and in how many seconds.

On the device

Real time in a phone browser with no server: privacy by design and near-zero cost per car.

Distribution and data

Inside Royal Motors after-sales, every handover adds a car; consented trip reports build a Tashkent near-miss dataset no outside team has.

A $14,900 pilot that pays back in about four months

Pilot, 3 months	USD
Two engineers (local rates)	9,000
20 phones, 100 windscreen mounts	3,400
Local test data with consent, 300 clips labelled	1,200
Hosting, CDN, store fees	300
Legal and privacy review	1,000
Total	**14,900**

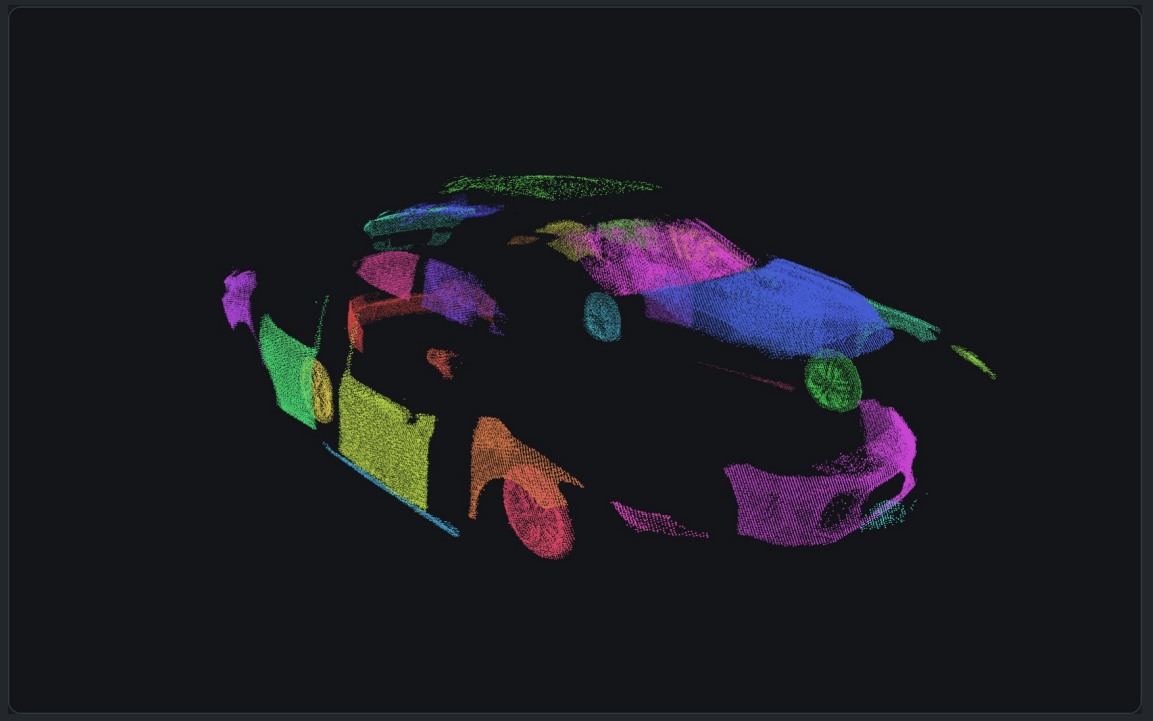
Per year after roll-out (assumptions)	USD
Safety Pack: $39 price − $13 direct cost = $26	
24,000 handovers and visits × 5% attach = 1,200 packs	31,200
Fleets: 300 vehicles × $3 per month	10,800
Contribution per year	**42,000**
Payback of the pilot	**≈ 4 months**

Every figure is our estimate, to be confirmed with Royal Motors in the pilot. Not counted: fewer crash repairs and downtime for fleets (IIHS: −27% rear-end crashes with a warning; we assume half until local data proves more) and insurer discounts. Running cost per car is near zero because inference runs on the phone.

Next product: one walk-around, the car in 3D, part by part



The Royal Motors test-drive Camry at the venue: 29 s video + 9 photos



Split into 32 named exterior parts, each sized and searchable

95 of 95 views placed. Scale from the published length; then wheelbase +2 mm, width +16 mm, height +4 mm vs published. Exported as glTF + JSON.

For after-sales: a 3D condition record at handover and trade-in, and parts lookup. Hidden parts would come from the Toyota/BYD catalogue by VIN.

Three months to a decision

Month 1

20 Royal Motors loaner and test-drive cars; record local footage with consent; tune on Tashkent streets, day and night.

Month 2

Safety Pack at handover in one Toyota and one BYD centre; one fleet customer with 50 vehicles.

Month 3

Measure attach rate, warnings per 100 km, driver feedback and incidents; decide on roll-out.

Our ask: one after-sales manager as pilot owner, 20 loaner cars, and access to one fleet customer.

